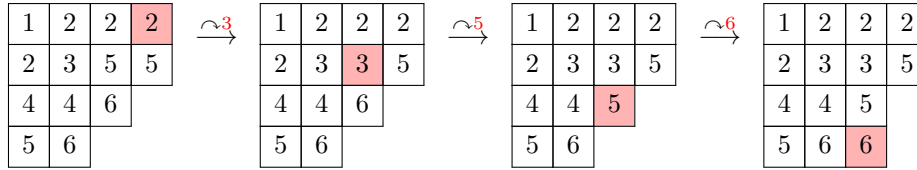


Problem 1.1. Compute the product of the following two SSYT both ways (insertion and rectification)

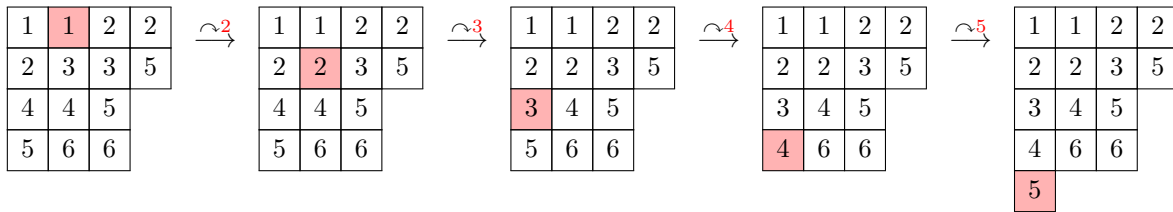
$$T = \begin{array}{|c|c|c|c|} \hline 1 & 2 & 2 & 3 \\ \hline 2 & 3 & 5 & 5 \\ \hline 4 & 4 & 6 & \\ \hline 5 & 6 & & \\ \hline \end{array} \text{ and } U = \begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & \\ \hline \end{array}$$

Proof. Insertion: $T \cdot U = ((T \leftarrow 2) \leftarrow 1) \leftarrow 3$

$(T \leftarrow 2)$



$(T \leftarrow 2) \leftarrow 1$



$((T \leftarrow 2) \leftarrow 1) \leftarrow 3 = T \cdot U$

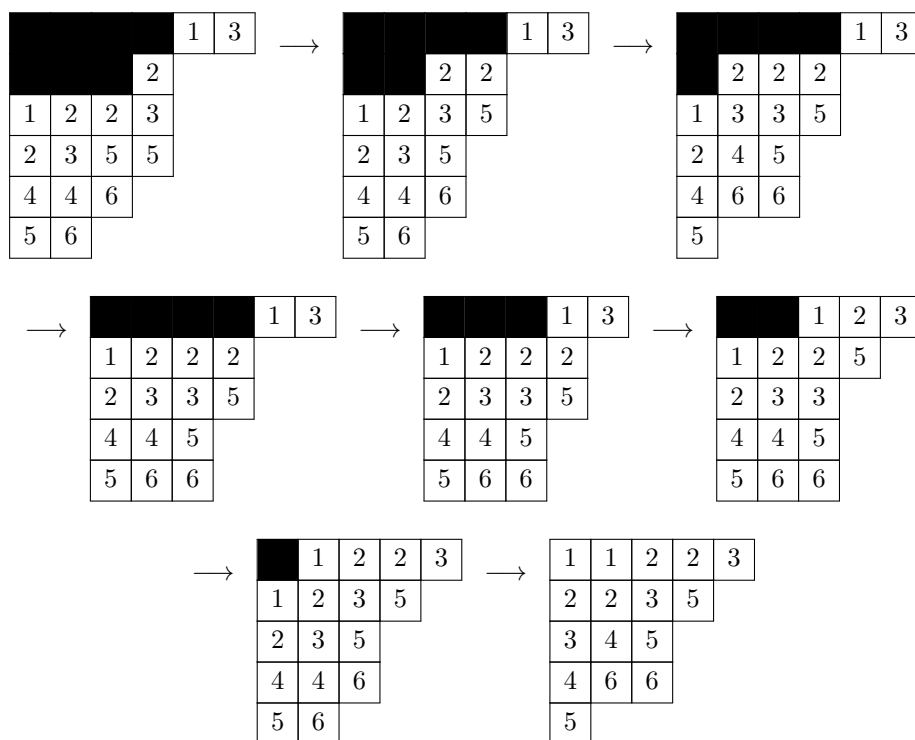
1	1	2	2	3
2	2	3	5	
3	4	5		
4	6	6		
5				

We do the same product by rectification on the next page.

- **Fulton—Chapter 1:** Exercises 1, 2 (pp. 15–16), compute product both ways
- **Fulton—Chapter 2:** Exercises 1, 2 (pp. 24–26)
- **Fulton—Chapter 3:** Exercises 1–3 (pp. 34–35)
- **Fulton—Chapter 4:** Exercises 12 (pp. 56) algorithm from class, 6,8,10,14 (pp. 52)
- **Fulton—Chapter 5:** Exercises 2 (pp. 66) 2,4,6
- **Fulton—Chapter 6:** Exercises 1,6 (pp. 74–76) 2,4,6 On 6 no need to deduce formula

$$T * U = \begin{array}{|c|c|c|c|c|} \hline & & & & 1 \ 3 \\ \hline & & & & 2 \\ \hline 1 & 2 & 2 & 3 & \\ \hline 2 & 3 & 5 & 5 & \\ \hline 4 & 4 & 6 & & \\ \hline 5 & 6 & & & \\ \hline \end{array}$$

We rectify this skew tableau



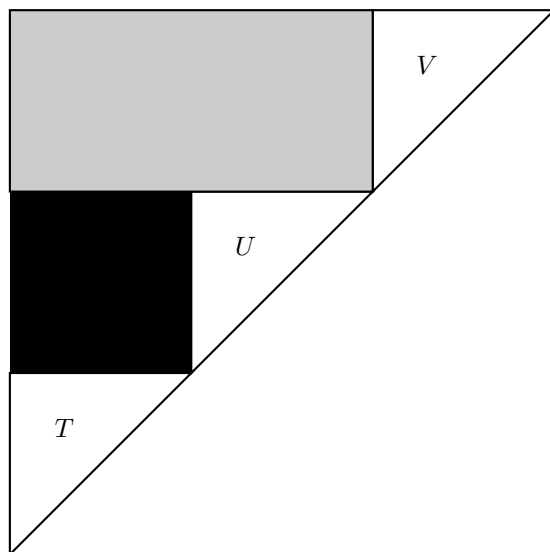
So we see that

$$\text{Rect}(T * U) = \begin{array}{|c|c|c|c|c|} \hline 1 & 1 & 2 & 2 & 3 \\ \hline 2 & 2 & 3 & 5 & \\ \hline 3 & 4 & 5 & & \\ \hline 4 & 6 & 6 & & \\ \hline 5 & & & & \\ \hline \end{array} = T \cdot U$$

□

Problem 1.2. Show that associativity of the product (Claim 1) $T \cdot U \cdot V$ follows from Claims 2 and 3 by forming the appropriate skew tableau $T * U * V$ from three given tableaux T, U, V .

Proof. Observe the structure of $T * U * V$



□